

KHUZIR, Russian FederationWMO: **306370**Lat: **53.190N**Lon: **107.330E**Elev: **462**StdP: **95.90**Time Zone: **8.00 (E08)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-31.3	-28.5	-34.1	0.2	-30.6	-31.6	0.2	-28.0	10.4	-14.4	9.6	-12.4	2.2	260	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	22.5	16.5	21.1	15.7	19.9	15.0	17.5	21.0	16.6	19.9	15.8	19.1	2.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	12.0	19.2	15.1	11.3	18.6	14.2	10.7	18.1	51.2	21.4	48.4	19.9	45.9	19.1	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.6	8.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-33.0	19.2	2.9	1.2	-35.1	20.1	-36.8	20.8	-38.4	21.5	-40.5	22.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.2	-17.7	-17.9	-9.2	-0.3	5.6	11.0	15.6	15.5	9.0	1.1	-6.2	-9.9
	DBStd	12.22	5.78	5.98	6.40	4.04	3.34	2.83	2.50	2.69	3.60	3.60	4.89	4.51
	HDD10.0	4150	857	782	596	308	143	21	1	2	61	276	487	618
	HDD18.3	6769	1116	1016	854	558	396	221	89	93	281	534	737	876
	CDD10.0	433	0	0	0	0	5	51	173	173	31	0	0	0
	CDD18.3	11	0	0	0	0	0	0	4	7	0	0	0	0
	CDH23.3	18	0	0	0	0	0	0	6	12	1	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	3.0	3.1	2.5	2.7	2.9	2.9	2.2	2.3	2.4	3.1	3.6	4.3	4.0
Precipitation	PrecAvg	356	12	8	7	14	27	52	74	73	39	16	17	16
	PrecMax	436	47	14	23	30	51	109	147	123	85	38	31	46
	PrecMin	280	3	1	1	4	7	21	39	32	14	2	7	5
	PrecStd	43	8	3	4	6	11	21	26	28	17	8	5	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.1	1.0	6.8	13.8	18.7	20.4	23.9	24.7	20.0	12.2	7.1	3.3
		MCWB	-5.3	-2.0	2.3	7.0	9.5	13.4	16.2	17.3	14.6	6.9	3.0	0.3
	2%	DB	-5.6	-2.6	4.1	10.2	15.2	18.7	22.4	22.9	17.9	10.1	4.8	-0.5
		MCWB	-7.3	-4.8	0.3	4.4	8.1	12.4	16.2	17.2	13.1	6.0	0.9	-3.0
	5%	DB	-7.3	-6.0	2.2	7.8	12.9	17.4	21.2	21.4	16.4	8.6	2.8	-2.1
		MCWB	-8.6	-7.7	-1.1	2.9	7.3	11.7	15.6	16.3	12.8	5.2	-0.4	-4.2
	10%	DB	-9.0	-8.8	0.6	5.8	11.0	16.0	20.1	20.2	15.1	7.2	0.8	-3.3
		MCWB	-10.3	-10.2	-2.2	1.8	6.0	11.1	15.1	15.6	11.7	3.8	-1.8	-5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.0	-1.8	2.2	7.2	10.8	15.2	18.2	19.1	15.4	8.0	3.5	0.2
		MCDB	-3.0	0.9	5.7	12.8	16.5	18.6	21.5	22.5	18.3	10.7	6.9	3.3
	2%	WB	-7.1	-4.8	0.5	5.0	8.9	13.7	17.2	18.0	14.4	6.9	1.2	-2.6
		MCDB	-5.8	-2.6	3.8	10.1	14.2	17.3	20.9	21.8	17.3	9.7	4.4	-0.7
	5%	WB	-8.5	-7.7	-0.9	3.2	7.6	12.5	16.4	17.0	13.1	5.5	-0.3	-4.1
		MCDB	-7.3	-6.1	2.1	7.2	12.4	16.0	20.2	20.2	15.8	8.4	2.4	-2.0
	10%	WB	-10.2	-10.2	-2.2	1.9	6.4	11.6	15.7	16.2	11.9	4.2	-1.6	-5.3
		MCDB	-9.1	-8.9	0.5	5.5	10.6	15.1	19.4	19.2	14.8	6.9	0.7	-3.5
Mean Daily Temperature Range	5% DB	MDBR	8.2	10.0	9.7	8.3	8.6	7.9	7.4	7.3	7.8	7.6	7.0	6.4
		MCDBR	9.4	11.1	10.1	10.4	11.7	9.9	8.7	8.8	9.1	9.9	8.9	8.0
		MCWBR	8.3	9.4	7.6	6.9	7.3	5.8	5.0	5.2	6.3	7.3	7.1	6.8
	5% WB	MCDBR	9.2	11.1	10.1	10.2	10.6	8.6	7.9	7.8	8.5	9.9	8.6	7.8
		MCWBR	8.3	9.5	7.7	7.0	6.9	6.1	5.5	5.3	6.3	7.7	7.0	6.8
Clear-Sky Solar Irradiance	taub	0.260	0.286	0.314	0.358	0.407	0.440	0.426	0.380	0.333	0.304	0.274	0.256	
	taud	2.105	2.096	2.152	2.301	2.212	2.213	2.303	2.413	2.491	2.430	2.171	2.068	
	Ebn at Noon	703	804	864	873	842	814	820	843	848	796	680	619	
	Edh at Noon	70	98	116	115	133	134	121	102	84	71	65	60	
All-Sky Solar Radiation	RadAvg	1.14	2.22	3.79	4.92	5.55	5.96	5.52	4.69	3.39	2.08	1.11	0.77	
	RadStd	0.06	0.09	0.15	0.33	0.38	0.35	0.39	0.20	0.28	0.18	0.07	0.06	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	-2.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page